

## Dynamic Dashboard Metrics Design (Enterprise-Level)

### 1. Numeric vs Numeric

Metrics: Correlation (r), Variance

Score =  $|r| * \sqrt{\text{Var}(x) * \text{Var}(y)}$

Chart: Scatter / Line

### 2. Categorical vs Numeric

Metrics: Group Mean, ANOVA

Score =  $\text{SS\_between} / \text{SS\_total}$

Chart: Bar

### 3. Categorical vs Categorical

Metrics: Chi-Square, Cramer's V

$V = \sqrt{\text{chi}^2 / (n * \min(k-1, r-1))}$

Chart: Stacked Bar / Heatmap

### 4. Time Series

Metrics: Slope (Regression), Std Dev

TrendScore =  $|\text{slope}| / \text{stdDev}$

Chart: Line

### 5. Column Profiling

Metrics: Null %, Unique Count, Cardinality

Used to filter useless columns

### 6. Final Score

FinalScore =  $0.5 * \text{RelationScore} + 0.3 * \text{ColumnScore} + 0.2 * \text{Coverage}$

### Implementation Flow

1. Fetch schema
2. Classify columns
3. Generate pairs
4. Compute metrics
5. Rank scores
6. Return top charts